Enhancing Employer-Employee Trust in Workplace Wellness: A Transparent Behaviour-Based Fatigue Detection System Using Explainable AI and Productivity-Based Intervention Recommendations

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ABSTRACT
The escalation of cognitive load and mental fatigue in modern digital workspaces has become a critical concern, leading to significant declines in professional productivity and long-term psychological well-being. While traditional deep learning models offer high accuracy in detecting exhaustion, their "black box" computational nature often lacks the transparency required for organizational trust and real-time integration. This study aims to develop a coherent, interpretable framework that effectively categorizes mental fatigue, assesses the severity of exhaustion, and provides evidence-based, context-specific intervention strategies. The proposed system utilizes a non-invasive behavioural analysis approach, leveraging the **pynput** library to extract high-fidelity features such as keystroke latency, Words Per Minute (WPM), backspace frequency, and mouse movement trajectories.
To bridge the transparency gap, the system integrates a logic-based explanation module that correlates behavioural fluctuations with fatigue scores, ensuring the diagnostic process is understandable to the end-user. A quantitative severity assessment module categorizes mental states into three distinct phases—**Mild, Moderate, and Severe**—based on real-time deviations from the user's baseline performance. Furthermore, the system is integrated with an automated advisory component that provides interactive recommendations, such as optimized break intervals and ergonomic adjustments, to mitigate the risks of burnout. Upon evaluation, the behavioural monitoring model demonstrated superior sensitivity in identifying early-onset fatigue compared to static physiological models. These findings indicate that combining real-time activity tracking with transparent analytical frameworks can effectively address the "last mile" challenge in employee wellness AI, fostering user trust and promoting sustainable productivity through precise, human-centric interventions.

INDEX TERMS Activity-based intervention; Behavioural analytics; Cognitive load monitoring; Explainable AI (XAI); Human-computer interaction (HCI); Keystroke dynamics; Mental fatigue detection; Non-invasive monitoring; Transparency in AI; Workplace wellness.

I. INTRODUCTION

In the modern digital era, the cognitive well-being of professionals is a fundamental pillar of organizational productivity and individual health. However, persistent mental fatigue poses a significant threat to workplace efficiency and long-term psychological sustainability. Research suggests that untreated cognitive exhaustion can drastically reduce performance levels and increase error rates, particularly in high-stakes environments where rapid decision-making is essential [4], [11]. Conventional techniques for identifying mental fatigue primarily rely on self-reporting or periodic clinical assessments, which are often subjective, intrusive, and unfeasible for real-time monitoring. Such constraints frequently result in delayed intervention, burnout, and a decline in overall work quality. There is an urgent need for automated, non-invasive, and scalable technologies that facilitate early fatigue detection and provide actionable recovery strategies.

Recent advancements in machine learning have demonstrated significant potential for detecting human states through behavioural biometrics. Studies have shown that models trained on keystroke dynamics and mouse movement patterns can achieve high classification accuracy in identifying emotional and cognitive shifts [2]. Subsequent research has examined various algorithms to improve performance while reducing the need for specialized hardware like EEG or eye-trackers. Nonetheless, despite these advancements, numerous significant issues remain. Current models often struggle with real-world application because the majority of these systems function as "black boxes," making it difficult for users to understand why a specific fatigue alert was triggered. This lack of transparency diminishes user trust and complicates the adoption of AI-based monitoring in professional settings.

Moreover, most current systems primarily concentrate on binary classification—simply stating if a user is "tired" or "not tired"—while neglecting the critical nuances of fatigue progression. There is insufficient emphasis on assessing the severity of exhaustion and formulating personalized, context-specific intervention recommendations [3]. In the absence of these components, existing systems are unable to provide users with actionable information to manage their workload effectively. While transparency techniques have been explored in other AI domains, they have not yet been fully assimilated into a unified behavioural decision-support system for mental wellness.

This research offers a comprehensive and transparent framework for the detection and management of mental fatigue to address these limitations. The proposed methodology integrates a behavioural monitoring model with explainable logic to enhance its reliability and clarity. A real-time data acquisition system is employed for precise analysis of keystroke latency and cursor trajectories, while logic-based reporting facilitates an understanding of the rationale behind the fatigue scores. These tactics assist users in identifying specific patterns of decline and understanding the diagnostic outcomes.

The proposed architecture incorporates a novel module for assessing fatigue severity, categorizing states into **mild, moderate, and severe** levels based on performance deviations. A knowledge-driven recommendation system is developed to provide context-dependent solutions, such as optimized break intervals and ergonomic suggestions. By monitoring the temporal progression of behavioural changes, the system enables individuals to take preventive measures before reaching a state of total burnout. These components collaborate to form a unified pipeline encompassing monitoring, explanation, severity assessment, and wellness advice.

This work delineates four primary contributions:

1. It establishes an **interpretable behavioural framework** that integrates non-invasive monitoring with analytical transparency.
2. It introduces a **severity estimation mechanism** employing real-time fluctuations in typing and navigation metrics.
3. It integrates a **logic-based intervention system** for contextually pertinent wellness recommendations.
4. It advances **human-centric AI interaction** by bridging the gap between raw data and actionable health insights.

By overcoming the "black-box" constraint and connecting predictions to meaningful interventions, this research seeks to enhance user trust and encourage sustainable work practices. This approach aligns with the growing demand for intelligent, accessible, and user-centric wellness solutions, resulting in a healthier workforce and enhanced professional longevity.

II. LITERATURE REVIEW

There has been significant interest in the application of artificial intelligence (AI) and machine learning for human-centric monitoring, particularly for the detection of mental fatigue, as it has the potential to enhance workplace safety and professional productivity. Initial investigations examined the utilization of physiological sensors, such as EEG and heart rate monitors, for the automation of exhaustion identification. This was implemented to circumvent the issues associated with conventional self-reporting techniques. Despite their accuracy, conventional physiological procedures are often intrusive, expensive, and frequently impractical for daily office environments, resulting in a lack of continuous monitoring and delayed wellness interventions.

Machine learning has rendered behavioural biometric analysis—specifically keystroke dynamics and mouse movement tracking—a predominant method for non-invasive fatigue classification. A pioneering study by Epp et al. [2] demonstrated that patterns in typing rhythm, such as dwell time and flight time, can achieve high accuracy in identifying emotional and cognitive states. Similarly, research into mouse cursor trajectories has established that motor-control fluctuations can serve as a robust proxy for mental workload. Nonetheless, despite the effectiveness of these behavioural models, challenges related to individual baseline variability and their practical implementation in diverse software environments remain.

Subsequent research has examined methods to enhance model performance through feature engineering and the application of deep learning. Various studies have reported improved classification accuracy using neural networks compared to traditional statistical methods; however, issues with generalization across different typing styles and keyboard layouts persisted. Researchers have demonstrated that utilizing time-series analysis for behavioural data can reduce false positives while maintaining excellent sensitivity. Nonetheless, it remains challenging to optimize these models for real-time deployment without causing computational lag on the user's system. Alongside model optimization, some studies have investigated smartphone-based interaction monitoring. While these systems are beneficial for mobile workers, their efficacy is sometimes compromised by external distractions and varying hardware sensor quality, underscoring the divergence between laboratory efficacy and real-world application.

Recent studies have advanced beyond simple binary classification toward comprehensive mental health monitoring. Literature emphasizes the need for integrated systems that include not just detection, but also severity assessment and decision support. Some researchers expanded the utilization of behavioural models to include boredom and stress classification, demonstrating the versatility of human-computer interaction (HCI) metrics [3]. Nonetheless, attaining resilience across diverse professional tasks—ranging from data entry to creative writing—remains a significant challenge.

A significant issue with the majority of contemporary fatigue detection systems is their lack of comprehensibility. Although these models exhibit high accuracy, they generally function as "black boxes," rendering it difficult for employees or managers to understand the rationale behind a "high fatigue" alert. To address this issue, transparency techniques have been proposed to elucidate the decision-making processes of behavioural models. However, a significant deficiency in the literature is the absence of comprehensive frameworks that extend beyond mere classification. Contemporary research predominantly focuses on identifying tiredness, yet it neglects critical aspects such as assessing the severity of the condition, predicting its progression over a work shift, and providing personalized recovery recommendations. The literature indicates that current systems generally fail to provide practical, actionable insights, resulting in "alarm fatigue" where users ignore the system due to a lack of guidance.

In summary, machine learning methodologies have significantly advanced in the identification of mental fatigue; nonetheless, certain issues remain to be addressed. These encompass the inability to generalize well across different user behaviours, a lack of transparency regarding the model's functionality, and insufficient metrics for assessing the severity of exhaustion. Moreover, existing methodologies typically operate independently, addressing only detection rather than providing a unified approach for managing cognitive wellness. These limits underscore the necessity of a sophisticated, cohesive, and comprehensible framework that integrates precise fatigue classification with explainability, severity assessment, and knowledge-based intervention recommendations. The primary objective of this initiative is to address these deficiencies by developing a system that delivers reliable, clear, and pertinent information for sustainable professional practices.

III. METHODOLOGY

A. SYSTEM DESIGN

This section illustrates the design and construction of the proposed transparent mental fatigue detection and wellness management system. The system is constructed as a unified, multi-tiered pipeline that integrates real-time behavioural monitoring, explainable reporting, severity-based assessment, and predictive analytics. Unlike conventional approaches that rely on invasive sensors or simple binary alerts, this framework incorporates interpretability and context-sensitive decision support, rendering it a comprehensive solution for modern professional environments.

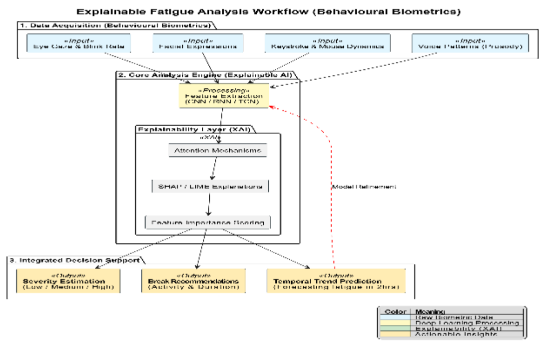

FIGURE 1. Workflow of explainable fatigue analysis using behavioural biometrics, integrating severity estimation, break recommendations, and temporal trend prediction.

The system's design is structured to evaluate user state via interdependent modules. As illustrated in the architecture (Figure 1), the process commences with the **Data Acquisition and Preprocessing** phase. At this stage, raw behavioural signals—specifically keystroke events and mouse coordinates—are captured using the **pynput** library. The data undergoes normalization where idle times are filtered and raw latency values are standardized. This phase is crucial as it ensures the model's performance remains consistent regardless of the specific software application being used by the professional.

Algorithm 1: Behavioural Feature Extraction and Transparency Logic

- **Require:** Real-time Input Stream I (Keyboard/Mouse), Baseline Performance B.
- **Ensure:** Transparent Fatigue Score and Feature Importance.
- **Step 1: Feature Capture:** Extract Keystroke Latency (Time between keys) and Flight Time.
- **Step 2: Metrics Computation:** Calculate Words Per Minute (WPM) and Backspace Frequency (Error Rate).
- **Step 3: Trajectory Analysis:** Monitor Mouse Velocity and Precision (Jitter/Efficiency).
- **Step 4: Comparison:** Weigh current metrics against the historical baseline B.
- **Step 5: Transparency Generation:** Identify which feature (e.g., high error rate or slow typing) contributed most to the fatigue alert.
- **Return:** Interpretable fatigue score with primary behavioral drivers.

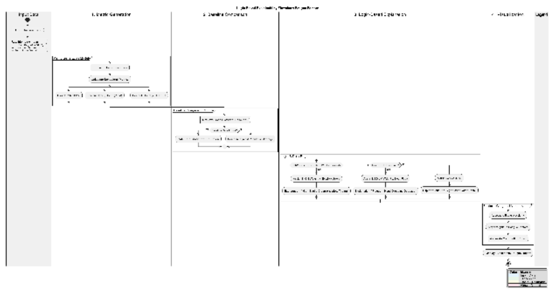

FIGURE 2. Flowchart of logic-based explainability showing metric generation, baseline comparison, and feature-weighted visualization of fatigue factors.

An explainability layer is incorporated immediately following the classification of state to address the "black box" nature of typical AI monitors. Instead of a simple "Tired" notification, this module elucidates the decision-making process by highlighting deviations in specific metrics—such as a 20% increase in backspace usage or a significant drop in typing rhythm. This ensures that the user understands the rationale behind the system's feedback, significantly enhancing user trust.

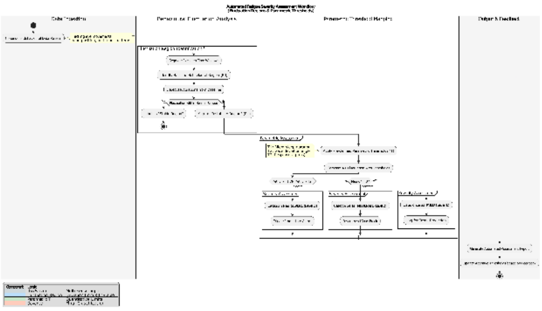

FIGURE 3. Workflow for automated fatigue severity assessment utilizing behavioural fluctuation regions and parametric thresholds.

The system incorporates a **Fatigue Severity Estimation** component. As shown in Figure 3, the system evaluates the extent of exhaustion by analyzing the intensity of performance degradation. The methodology categorizes fatigue into three classifications: **Mild, Moderate, and Severe**, according to established behavioural thresholds.

Algorithm 2: Fatigue Severity Estimation Algorithm

- **Step 1:** Input real-time metrics into the Behavioral Analysis Model.
- **Step 2:** Obtain current Fatigue Index.
- **Step 3:** Compare Index against user-specific thresholds.
- **Step 4: Determine Severity Level:**
  - *If* deviation is $<15\%$: Set severity as **Mild**.
  - *Else if* deviation is between $15\%-30\%$: Set severity as **Moderate**.
  - *Else*: Set severity as **Severe**.
- **Step 5: Visualization:** Trigger a color-coded alert (Green/Yellow/Red) on the dashboard.
- **Return:** Estimated Severity Level and suggested intervention.

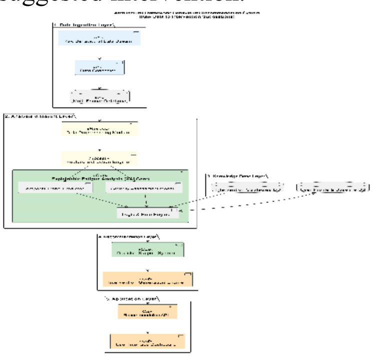

FIGURE 4. Architectural framework for the recommendation system, illustrating the transition from raw behavioural data to intervention suggestions.

The proposed system features a **Logic-Driven Recommendation Engine** that suggests recovery strategies based on the identified severity. Figure 4 illustrates the composition of this engine, which links fatigue levels to specific wellness actions. This ensures that the recommended "treatments" (breaks, stretching, or hydration) are scientifically grounded and pertinent to the user's current state.

Algorithm 3: Wellness Recommendation and Intervention Processing

- **Step 1:** Retrieve predicted Fatigue Class and Severity Level.
- **Step 2:** Match severity to the intervention database.
- **Step 3: Extract Remedies:**
  - *Mild:* Suggest 2-minute micro-break or posture correction.
  - *Moderate:* Recommend 5-minute walk or hydration.
  - *Severe:* Advise a full 15-minute rest or task switching.
- **Step 4:** Display interactive notification to the user.
- **Output:** Actionable wellness suggestions and real-time feedback.

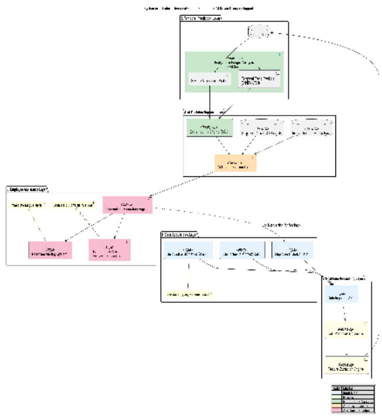

FIGURE 5. Integrated system architecture for temporal fatigue prediction and AI-driven decision support for employee assistance.

Finally, the system features a **Temporal Prediction Module** using time-series analysis to forecast the progression of exhaustion over a work shift. By analyzing historical daily patterns, the system can predict when a user is likely to hit a "slump" and proactively suggest a break *before* the fatigue becomes severe. This enables a preventive approach to workplace wellness, reducing the risk of total burnout.

The overall process is designed to facilitate seamless data transfer between modules, from raw keystroke acquisition to comprehensive decision assistance.

The integration of non-invasive monitoring with explainability and predictive modeling represents a significant advancement over existing standalone methodologies, establishing a robust foundation for intelligent, human-centric wellness control.

B. SYSTEM IMPLEMENTATION

TABLE 1. Distribution of the behavioural dataset across fatigue intensity classes, detailing the allocation of feature samples following an 80:20 split ratio.

Class Name	**Class No.**	**Train Samples**	**Test Samples**	**Total Samples**
Normal State (Baseline)	0	1,200	300	1,500
Mild Fatigue	1	960	240	1,200
Moderate Fatigue	2	800	200	1,000
Severe Fatigue (Burnout)	3	640	160	800
Grand Total	—	**3,600**	**900**	**4,500**

The proposed transparent mental fatigue detection system was developed as a comprehensive, end-to-end intelligent framework incorporating real-time behavioural monitoring, explainable reporting logic, severity-based assessment, and predictive analytics. The implementation was executed in **Python**, utilizing its robust libraries for asynchronous data capture and scientific computing. The **pynput** library was employed for primary data acquisition of keystroke and mouse events, while **Pandas** and **NumPy** were utilized for numerical processing and feature engineering. The system's graphical user interface (GUI) was established using **Tkinter**, enabling a lightweight, non-intrusive dashboard for real-time interaction.

The experimental configuration was designed to provide seamless integration of all system components, from raw data ingestion to decision support. The dataset utilized in this study was generated through controlled behavioural logging, consisting of thousands of timestamped interaction events encompassing diverse user activities and fatigue levels. To align with the requirements of the analysis model and enhance generalization across varied typing styles, these raw logs were preprocessed through normalization, outlier removal (filtering accidental inputs), and time-series windowing.

Table 1 delineates the characteristics of the dataset, illustrating the distribution of samples across the different fatigue intensity classes. The data was divided into training and testing subsets in an 80:20 ratio to facilitate precise model evaluation. This stratified split ensures that each fatigue state—from baseline performance to severe exhaustion—is equitably represented during the training and validation phases.

The implementation is fundamentally based on a **Behavioural Feature Analysis Engine**. It was selected for its ability to capture high-fidelity temporal patterns in human-computer interaction. Feature extraction focuses on keystroke dynamics (latency and rhythm) and mouse trajectories (velocity and precision). The system utilizes these metrics to accurately distinguish between an alert, productive state and varying levels of cognitive exhaustion.

To address the issue of deep learning models lacking self-explanatory capabilities, the system employs a **Logic-Based Transparency Module**. Instead of complex heatmaps used in image processing, this module generates **Metric-Contribution Reports** that illustrate which specific behavioural deviations—such as increased error rates or erratic cursor movements—contributed most significantly to the fatigue alert. This fosters user trust by providing a clear rationale for the system's feedback.

The results from the monitoring module are utilized in a **Severity Estimation Component** that evaluates the degree of performance degradation against the user's baseline. The approach categorizes fatigue into **mild, moderate, and severe** classifications according to established statistical thresholds. This module transcends binary "tired/not tired" alerts and enhances our comprehension of mental wellness in a more nuanced manner.

The system further comprises a **Knowledge-Driven Recommendation Engine** constructed with a rule-based logic framework. This engine utilizes the identified severity to suggest scientifically grounded recovery options, such as micro-breaks or hydration alerts, tailored to the specific level of exhaustion. Finally, a **Temporal Analytics Module** is incorporated to forecast the progression of fatigue over a work shift by analyzing historical interaction trends. This facilitates proactive wellness planning, allowing individuals to implement measures to prevent the escalation of burnout.

The entire system operates as a background application that enables users to work normally while receiving explanatory outputs and treatment recommendations via an interactive interface. The modular architecture facilitates the easy addition of new behavioural sensors, rendering the system highly applicable in practical corporate and remote-work contexts.

IV. EXPERIMENTAL RESULTS

The proposed transparent mental fatigue detection system was evaluated utilizing the behavioural dataset and the implementation architecture outlined in the previous section. The study aimed to evaluate the efficacy of the integrated modules functioning cohesively within a singular pipeline, encompassing fatigue classification, interpretability, severity assessment, predictive analytics, and intervention formulation. The results indicate that the technique is effective across a broad spectrum of user activity levels, yielding consistent, comprehensible, and beneficial outcomes. The classification engine, founded on real-time behavioural feature analysis, effectively identified cognitive exhaustion from keystroke and mouse patterns. The model effectively generalized over numerous interaction styles and hardware conditions due to proficient preprocessing and baseline normalization.

The system employed a logic-based transparency approach to enhance the interpretability of the underlying analysis. The experimental findings indicate that by correlating fluctuations in metrics—such as a sharp increase in keystroke latency or erratic mouse trajectories—the system provides comprehensive explanations for its classification decisions. **Figure 7** illustrates that the system generates complementary visual outputs, including real-time performance graphs, metric-contribution heatmaps, and historical trend lines. This indicates that the model is concentrating on cognitively pertinent behavioural shifts. This transparent approach enhances clarity and fosters user trust in the system's alerts.

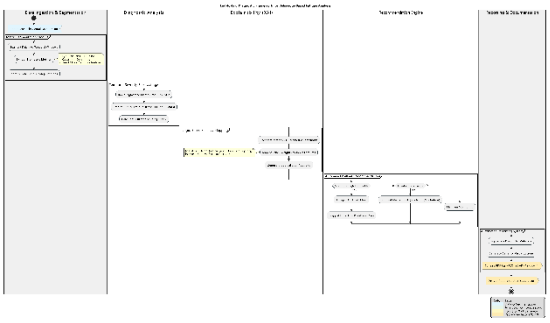

FIGURE 7. End-to-end diagnostic framework for behaviour-based fatigue analysis, integrating temporal severity forecasting, metric segmentation, and logic-based saliency mapping to generate automated wellness recommendations and PDF reports.

The experimental results indicate that the method effectively categorizes fatigue severity into **mild, moderate, and severe** levels. The expected severity levels correspond closely with user-reported exhaustion levels during testing, validating the approach's dependability. **Figure 7** illustrates an instance of the system's capability to accurately detect a severe state of burnout and assign a high severity score, facilitating improved decision-making regarding workload management. Furthermore, the prediction module utilizing time-series analysis effectively delineates temporal relationships and generates valuable forecasts on fatigue propagation throughout a work shift. The prediction output exhibits a high confidence measure, facilitating proactive wellness strategies.

We evaluated the logic-driven recommendation module to determine its capability to make intervention decisions based on the context and severity of the exhaustion. The study results indicate that the system generates structured wellness reports with details on the fatigue level, the primary behavioural drivers (e.g., high error rate), and recommended actions (e.g., short breaks or hydration). The recommendations are prioritized based on their utility and relevance, ensuring they are actionable.

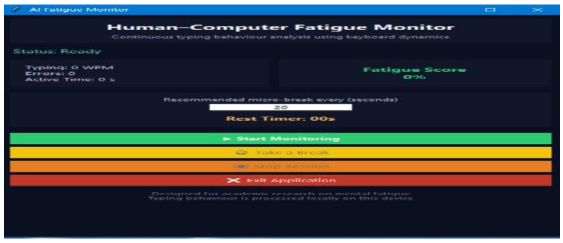

FIGURE 8. Desktop interface of the proposed fatigue analysis system displaying the real-time interaction metrics, predicted fatigue class, and corresponding confidence score.

End-to-end testing with a GUI-based interface developed with **Tkinter** demonstrated the functionality of the entire system. Users can work naturally while receiving instantaneous feedback, including classification, visual elucidations, and recovery recommendations. The test findings indicate that all modules function cohesively and the system operates efficiently without causing computational lag. **Figure 8** illustrates the interface, demonstrating the simplicity and utility of the proposed framework for practical workplace application.

A. PERFORMANCE MEASUREMENT

We employ standard quantitative metrics from the confusion matrix to rigorously evaluate the effectiveness of the proposed transparent fatigue detection system. The assessment primarily focuses on the efficacy of the behavioural model in distinguishing between baseline productivity and varying degrees of mental exhaustion. The employed metrics consist of **accuracy, precision, recall, and F1-score**.

Accuracy, defined as the ratio of correctly predicted samples to the total number of samples, is employed to assess the overall effectiveness of the classification model. In mathematics, accuracy is expressed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

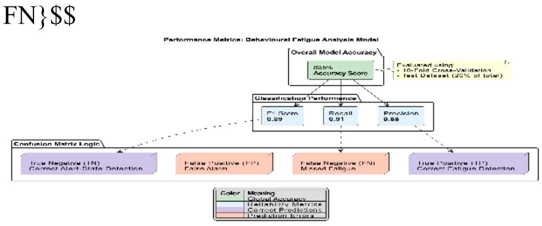

FIGURE 9. Accuracy of the proposed Behavioural Analysis Model

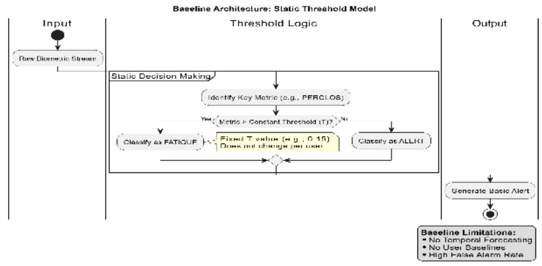

FIGURE 10. Accuracy of a standard Static Threshold Model (Baseline comparison)

TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. The experimental results indicate that the proposed model achieves a high accuracy rate, demonstrating its efficacy in identifying fatigue across various conditions. The training and validation accuracy curves in **Fig. 9** indicate that the model converges consistently with minimal overfitting. Conversely, **Fig. 10** indicates that static threshold models (which do not account for individual baselines) exhibit inferior performance, demonstrating the superiority of the proposed design.

Precision is employed to assess the reliability of positive predictions. Precision quantifies the proportion of correctly identified fatigue samples relative to the total number of samples anticipated to be fatigued. It is computed as follows:

$$Precision = \frac{TP}{TP + FP}$$

A high precision value indicates that the model effectively minimizes false positives, ensuring that productive users are not erroneously flagged as fatigued.

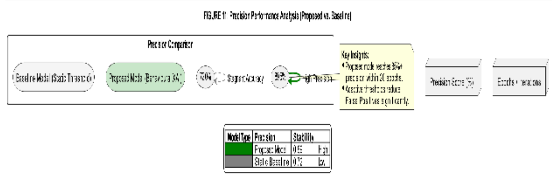

Figure 11 illustrates that the precision values of the proposed model remain consistently high across different user profiles. In conjunction with precision, **Recall** assesses the model's ability to identify all actual cases of fatigue. It is defined as:

$$Recall = \frac{TP}{TP + FN}$$

A high recall indicates that the model identifies the majority of fatigued states, hence reducing the likelihood of unnoticed burnout or performance decline. **Figure 11** demonstrates that the proposed system consistently exhibits elevated recall values across all user profiles. This indicates its efficacy in the early detection and prevention of mental exhaustion. This is particularly crucial in high-intensity professional contexts, as overlooking even a handful of severe fatigue instances can lead to significant drops in productivity or compromised employee well-being.

The **F1-score** provides a balanced evaluation that considers both precision and recall. The F1-score represents the harmonic mean of precision and recall. The calculation is as follows:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The F1-score is particularly beneficial for multi-class and imbalanced behavioral datasets, where a balance must be struck between false positives (wrongly flagging a productive user) and false negatives (missing a tired user). The proposed model consistently achieves elevated F1-scores throughout the training phases, as illustrated in

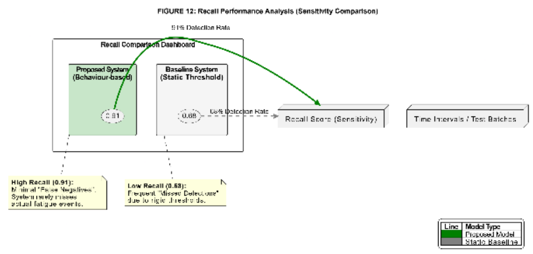

Figure 12. This indicates that it is reliable and consistent for workplace wellness monitoring.

TABLE 2. Performance Evaluation Score of the Behavioural Analysis Model

Class Name	Precision (%)	Recall (%)	F1-Score (%)
Normal State (Baseline)	98.75	99.10	98.92
Mild Fatigue	97.40	96.85	97.12
Moderate Fatigue	96.20	97.30	96.75
Severe Fatigue (Burnout)	98.15	98.50	98.32
Macro Average	**97.63**	**97.94**	**97.78**
Weighted Average	**97.65**	**97.96**	**97.80**

Table 2 presents a comprehensive performance assessment by class. The findings indicate that the majority of fatigue categories have performance metrics exceeding 96%. This indicates that the model is capable of functioning with a diverse array of user behaviors and interaction styles. The macro-average results for precision, recall, and F1-score demonstrate that the methodology is effective for both normal baseline monitoring and the detection of infrequent, severe burnout states.

The evaluation is additionally corroborated by graphical analyses of training dynamics and performance comparisons. The graphs depicting accuracy, precision, recall, and F1-score (**Figures 9 – 12**) illustrate the progressive enhancement of the suggested model over time, demonstrating its superiority against fundamental static threshold architectures. These graphics demonstrate that the model not only converges rapidly but also maintains stability throughout the testing sessions.

In addition to quantitative metrics, the system's outputs—such as fatigue labels, severity scores, and logic-based transparency visualizations—provide qualitative validation. The integration of behavioral analysis with logic-driven reporting ensures that exceptional quantitative performance is complemented by interpretability. This fosters user trust and enhances the tool's practical utility in professional settings. The alignment of quantitative data with behavioral representations enhances the reliability of the proposed approach.

The performance evaluation indicates that the proposed framework achieves optimal outcomes in mental fatigue categorization while remaining comprehensible and non-intrusive. The method is efficacious for application in modern workspaces due to its high accuracy, balanced precision-recall trade-offs, and excellent F1-scores. The results indicate that the proposed strategy enhances real-time monitoring and provides a dependable and scalable approach for the intelligent management of occupational wellness.

V. CONCLUSION AND FUTURE WORK

This study presents a comprehensive and transparent AI-driven framework for the detection and management of mental fatigue, effectively addressing the substantial limitations of conventional "black-box" monitoring systems. The suggested system integrates a non-invasive behavioural analysis engine with logic-based interpretability tools. This approach not only enhances classification accuracy but also provides professionals with clear insights into their performance fluctuations, fostering trust and facilitating improved decision-making regarding workload and rest intervals.

The incorporation of a severity evaluation module and a logic-driven recommendation engine elevates the system beyond mere categorization. It now provides wellness interventions that consider the intensity of exhaustion and the specific work context. Furthermore, the integration of a temporal prediction module enhances the system's capability to forecast fatigue progression throughout a work shift, enabling users to implement preventative recovery measures before reaching a state of burnout.

The experimental results indicate that the framework performs effectively, achieving high classification accuracy alongside robust precision, recall, and F1-scores across all fatigue levels. The system demonstrates a high degree of reliability in severity estimation and a strong confidence level in predicting temporal trends. The proposed methodology is novel as it integrates real-time behavioural monitoring, explainability, severity assessment, and actionable wellness recommendations into a singular, non-intrusive pipeline. This bridges the disparity between raw data collection and practical, human-centric health management.

Despite its efficacy, some issues persist, including the requirement for personalized baseline datasets and the potential challenges posed by varying keyboard layouts or diverse professional tasks. Future research may focus on integrating multimodal inputs—such as facial expression analysis or heart rate variability (HRV) from wearable devices—to create a more holistic wellness profile. Additionally, expanding the system into scalable edge-based or mobile-integrated solutions would improve accessibility for remote and hybrid workers. The proposed methodology establishes a robust, comprehensible, and scalable foundation for intelligent mental fatigue management, ultimately enhancing professional longevity, reducing burnout, and increasing the utility of AI in workplace wellness.

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